

 poolside/**Laguna-XS.2-INT4** 

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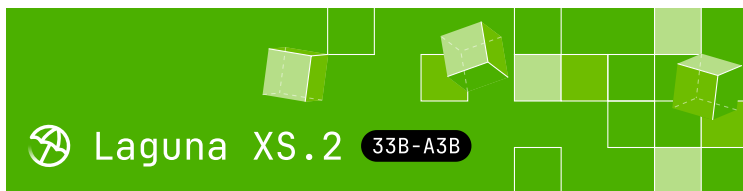
 Text Generation
  Safetensors
  vllm
  laguna
  laguna-xs.2
  conversational
  custom_code

 compressed-tensors
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 **Model card**
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Laguna XS.2-INT4

Laguna XS.2-INT4 is a 33B total parameter Mixture-of-Experts model with 3B activated parameters per token designed for agentic coding and long-horizon work on a local machine. It uses Sliding Window Attention with per-head gating in 30 out of 40 layers for fast inference and low KV cache requirements.

This is the INT4 variant with an FP8-quantized KV cache. The [BF16](#), [FP8](#) and [NVFP4](#) variants are also available on Hugging Face.

Highlights

Downloads last month
1,116



 **Safetensors** 

Model size 41B params

Tensor type F32 · I32 · BF16

 Chat template

 Files info

 **Inference Providers** **NEW**

 Text Generation

This model isn't deployed by any Inference Provider.

 Ask for provider support

 **Model tree for poolside/Laguna-X...**

Base model poolside/Laguna-XS.2

 Quantized (15) [this model](#)

 **Collection including poolside/Lag...**

Laguna XS.2  Collection

Designed for ... • 5 items • U... •  28

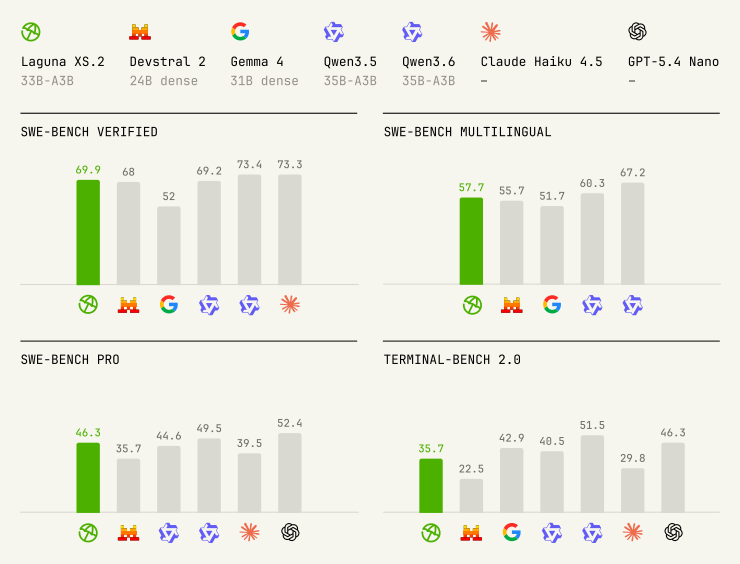
- **Mixed SWA and global attention layout:**
Laguna XS.2 uses sigmoid gating with per-layer rotary scales, enabling mixed SWA (Sliding Window Attention) and global attention layers in a 3:1 ratio (across 40 total layers)
 - **KV cache in FP8:** KV cache quantized to FP8, reducing memory per token
 - **Native reasoning support:** Interleaved thinking between tool calls with support for enabling and disabling thinking per-request
 - **Local-ready:** At 33B total parameters and 3B activated, Laguna XS.2 is compact enough to run on a Mac with 36 GB of RAM. [Available on Ollama](#)
 - **Apache 2.0 license:** Use and modify freely for commercial and non-commercial purposes
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Model overview

- Training: pre-training, post-training and reinforcement learning stages
- Number of parameters: 33B total with 3B activated per token
- Optimizer: Muon
- Layers: 40 layers (10 layers with global attention, 30 layers with sliding window attention)
- Experts: 256 experts with 1 shared expert
- Sliding Window: 512 tokens
- Modality: text-to-text

- Context window:262,144 tokens
- Reasoning support:interleaved thinking with preserved thinking

Benchmark results



Model	Size (total params.)	SWE-bench Verified	SWE-bench Multilingual	SWE-bench Pro (Public Dataset)
Laguna XS.2 (BF16)	33B	69.9%	57.7%	46.3%
Devstral Small 2	24B dense	68.0%	55.7%	-
Gemma 4 31B IT	31B dense	52.0%	51.7%	35.7%
Qwen3.5-35B-A3B	35B	69.2%	60.3%	44.6%
Qwen3.6-35B-A3B	35B	73.4%	67.2%	49.5%

Model	Size (total params.)	SWE- bench Verified	SWE- bench Multilingual	SWE- bench Pro (Public Dat ase
Claude Haiku 4.5	-	73.3%	-	39.5%
GPT-5.4 Nano	-	-	-	52.4%

We used the highest publicly-referenced scores for all comparison models across each benchmark. In almost all cases these were official scores published in release blog posts or equivalent, with the exception of Gemma 4 31B IT where the highest published scores were reported by the Qwen team and Claude Haiku 4.5 where the highest published (verified) scores for SWE- bench Pro and Terminal-Bench 2.0 are from their respective official leaderboards.

► Expand for benchmarking methodology

Usage

Laguna XS.2-INT4 has launch-day support in vLLM and Transformers.

The fastest way to get started is with our API, directly or using OpenRouter.

For complete usage instructions, see the main [Laguna XS.2 model card](#).

Local deployment

Laguna XS.2-INT4 is supported in vLLM and Transformers. Use Laguna-XS.2 with Ollama (with MLX support) and the mlx-lm framework for the best experience on your local machine.

vLLM

The full vLLM recipe is on the main [Laguna XS.2 model card](#) and on the [vLLM recipes page](#).

Quantization is detected automatically from `quantization_config` in this checkpoint, so the same command works with `poolside/Laguna-XS.2-INT4` substituted for the model ID. No extra flags required.

INT4 support landed in vLLM with [vllm-project/vllm#47154](#). This checkpoint mixes INT4 and group-quantized INT8 experts, which earlier builds rejected in the Marlin WNA16 MoE path. Use a vLLM build that includes this fix.

The FP8-quantized KV cache requires vLLM $\geq$ 0.22.0. Earlier versions produce scrambled output on non-Hopper GPUs because of a per-layer attention-head count bug, fixed in [vllm#42650](#). On older vLLM, disable the FP8 KV cache by adding `--kv-cache-dtype-skip-layers $(seq 0 39)`.

Transformers

The full Transformers recipe is on the main [Laguna XS.2 model card](#). Substitute `poolside/Laguna-XS.2-INT4` for the model ID; quantization is detected automatically from `quantization_config`.

Ollama

Visit [Ollama's model library](#) to pull to your local machine.

Controlling reasoning

Laguna XS.2 has native reasoning support and is designed to work best with *preserved thinking*, where reasoning content from prior assistant messages is preserved in the message history. This model will generally reason before calling tools and between tool calls.

► Expand for example

Disabling reasoning

You can disable thinking by setting `enable_thinking` to `False` in a request or by not providing `--default-chat-template-kwarg` `{"enable_thinking": True}` or equivalent when starting the server.

► Expand for example

For agentic coding use cases, we recommend enabling thinking and preserving reasoning in message history as outlined in the [Controlling reasoning] section.

License

This model is licensed under the [Apache 2.0 License](#).

Intended and Responsible Use

Laguna XS.2-INT4 is designed for software engineering and agentic coding use cases, and you

are responsible for confirming that it is appropriate for your intended application. Laguna XS.2-INT4 is subject to the [Apache 2.0 License](#), and should be used consistently with Poolside's [Acceptable Use Policy](#). We advise against circumventing Laguna XS.2-INT4 safety guardrails without implementing

ivalent mitigations appropriate for your use case.

Please report security vulnerabilities or safety concerns to security@poolside.ai.